

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	Planets / masses orbit in ellipses (with the Sun at a focus) (1) [Line from planet to Sun] sweeps equal area(s) in equal time(s) or sweeps area at a constant rate (1) Sun at a focus & line from planet to Sun (or radius vector) (1)	3			3		
		(ii)	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ or $mr\omega^2$ (1) Substitution of $\omega = \frac{2\pi}{T}$ or equivalent i.e. $v = \frac{2\pi r}{T}$ (1) Intermediate step seen e.g. $\frac{GMm}{r^2} = m \frac{4\pi^2}{T^2} r$ (1) Alternative: $T = 2\pi \sqrt{\frac{d^3}{G(M_1 + M_2)}}$ (1) $M_2 = 0$ or implied (1) $d = r$ and $M_1 = M_E$ (1)	3			3	3	
	(b)		Period of geostationary = 1 day (24 hr etc.) (1) Use of Kepler e.g. $\frac{T_1^2}{r_1^3} = \frac{T_2^2}{r_2^3}$ [accept substitution with 5.97×10^{24}] (1) Correct answer = 27.2 [days] (2.35×10^6 s) (1) Alternative: Substitution into period equation (1) Algebra (1) Correct answer = 27.2 [days] (2.32×10^6 s) (1)	1 1	1		3	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Assumption - Radius cannot be less than r_E (or inside Earth's surface etc.) or ignore mountains/air resistance accept no atmosphere or Earth is totally smooth (1) Use of Kepler or accept equation use again (1) Correct answer = 1.42 h, $\frac{1}{16.9}$ of a day, 5 100 s etc. (accept slightly larger due to, e.g. $r_E = 6\,470\text{ km}$ to be above atmosphere etc. which gives 1.45 h) (1)		3		3	2	
			Question 3 total	8	4	0	12	7	0